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Electric submersible pump (ESP) gas slug processor and mitigation system

Abstract

A downhole gas separator assembly. The gas separator comprises a drive shaft; a first fluid mover mechanically coupled to the drive shaft having a fluid inlet and a fluid outlet; a fluid reservoir concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the fluid reservoir and an outside surface of the drive shaft define a first annulus that is fluidically coupled to the fluid outlet of the first fluid mover; a second fluid mover having a fluid inlet and a fluid outlet, wherein the second fluid mover is located downstream of the fluid reservoir, and wherein the fluid inlet of the second fluid mover is fluidically coupled to the first annulus; and a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3195661	12/1964	Jackson et al.	N/A	N/A
4187912	12/1979	Cramer	N/A	N/A
4481020	12/1983	Lee et al.	N/A	N/A
4901413	12/1989	Cotherman et al.	N/A	N/A
4913630	12/1989	Cotherman et al.	N/A	N/A
5207810	12/1992	Sheth	N/A	N/A
5431228	12/1994	Weingarten et al.	N/A	N/A
5525146	12/1995	Straub	N/A	N/A
6066193	12/1999	Lee	N/A	N/A
6113675	12/1999	Branstetter	N/A	N/A
6155345	12/1999	Lee et al.	N/A	N/A
6382317	12/2001	Cobb	N/A	N/A
6412562	12/2001	Shaw	N/A	N/A
6705402	12/2003	Proctor	N/A	N/A
7461692	12/2007	Wang	N/A	N/A
7462225	12/2007	Ketter et al.	N/A	N/A
7766081	12/2009	Brown et al.	N/A	N/A
8079418	12/2010	Brown et al.	N/A	N/A
8955598	12/2014	Brown et al.	N/A	N/A
9283497	12/2015	Wang et al.	N/A	N/A
10378322	12/2018	Ejim et al.	N/A	N/A
10408035	12/2018	Kennedy et al.	N/A	N/A
11149535	12/2020	Brown et al.	N/A	N/A
11624269	12/2022	Sheth et al.	N/A	N/A
11661828	12/2022	Conrad et al.	N/A	N/A
11867035	12/2023	Brown et al.	N/A	N/A
11946472	12/2023	Sheth et al.	N/A	N/A
11965402	12/2023	Brown et al.	N/A	N/A
12000258	12/2023	Brown	N/A	E21B 43/38
12024990	12/2023	Brown et al.	N/A	N/A
12060780	12/2023	Sheth	N/A	E21B 43/128
2002/0153141	12/2001	Hartman et al.	N/A	N/A
2002/0174982	12/2001	Weingarten	N/A	N/A

2003/0111230	12/2002	Olson et al.	N/A	N/A
2004/0031608	12/2003	Hall et al.	N/A	N/A
2009/0044953	12/2008	Sheth et al.	N/A	N/A
2009/0065202	12/2008	Brown et al.	N/A	N/A
2009/0194295	12/2008	Thompson et al.	N/A	N/A
2010/0147514	12/2009	Swaringin	N/A	N/A
2010/0319926	12/2009	Reid	N/A	N/A
2011/0162832	12/2010	Reid	N/A	N/A
2012/0012332	12/2011	Rooks	N/A	N/A
2012/0073800	12/2011	Janitz	N/A	N/A
2013/0068455	12/2012	Brown et al.	N/A	N/A
2014/0174734	12/2013	Gill et al.	N/A	N/A
2015/0192141	12/2014	Nowitzki et al.	N/A	N/A
2016/0130922	12/2015	Wilson et al.	N/A	N/A
2016/0130923	12/2015	Nowitzki et al.	N/A	N/A
2016/0201444	12/2015	Hardee et al.	N/A	N/A
2016/0222770	12/2015	Kirk et al.	N/A	N/A
2016/0312590	12/2015	Chang et al.	N/A	N/A
2017/0081952	12/2016	Osborne et al.	N/A	N/A
2018/0274344	12/2017	Ejim et al.	N/A	N/A
2019/0162063	12/2018	Brown	N/A	F04D 13/10
2019/0234193	12/2018	Fehri	N/A	N/A
2019/0264551	12/2018	Brown et al.	N/A	N/A
2019/0316452	12/2018	Brown et al.	N/A	N/A
2020/0141223	12/2019	Brown et al.	N/A	N/A
2020/0362649	12/2019	Brown et al.	N/A	N/A
2022/0065090	12/2021	Ejim	N/A	N/A
2023/0010704	12/2022	Sheth et al.	N/A	N/A
2023/0014297	12/2022	Brown et al.	N/A	N/A
2023/0105823	12/2022	Brown et al.	N/A	N/A
2023/0111850	12/2022	Sheth et al.	N/A	N/A
2023/0220751	12/2022	Sheth et al.	N/A	N/A
2024/0102366	12/2023	Brown et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106076006	12/2015	CN	N/A
205965282	12/2016	CN	N/A
2017053624	12/2016	WO	N/A
2020072078	12/2019	WO	N/A
2021177981	12/2020	WO	N/A
2023282919	12/2022	WO	N/A
2023282920	12/2022	WO	N/A
2023215027	12/2022	WO	N/A
2024072509	12/2023	WO	N/A
2024072510	12/2023	WO	N/A

OTHER PUBLICATIONS

Schlumberger Oilfield Glossary entry for “slug flow”, accessed Sep. 8, 2023 via glossary.slb.com. cited by applicant

Electronic Acknowledgment Receipt, Specification and Drawings for International Application No. PCT/US2021/042389, entitled “Integrated Gas Separator and Pump,” filed Jul. 20, 2021, 50 pages. cited by applicant

Electronic Acknowledgment Receipt, Specification and Drawings for International Application No. PCT/US2021/042395, entitled “Electric Submersible Pump (ESP) Gas Slug Processor and Mitigation System,” filed Jul. 20, 2021, 51 pages. cited by applicant

Foreign Communication from Related Application—International Search Report and Written Opinion of the International Searching Authority, International Application No. PCT/US2020/022038, dated Nov. 27, 2020, 13 pages. cited by applicant

Foreign Communication from Related Application—International Search Report and Written Opinion of the International Searching Authority, International Application No. PCT/US2021/042395, dated Mar. 25, 2022, 14 pages. cited by applicant

Foreign Communication from Related Application—International Search Report and Written Opinion of the International Searching Authority, International Application No. PCT/US2021/042389, dated Apr. 4, 2022, 11 pages. cited by applicant

Foreign Communication from Related Application—International Search Report and Written Opinion of the International Searching Authority, International Application No. PCT/US2023/013758, dated Jun. 9, 2023, 10 pages. cited by applicant

Filing Receipt, Specification and Drawings for U.S. Appl. No. 17/954,855, filed Sep. 28, 2022, entitled “Electric Submersible Pump (ESP) Assembly Fluid Intake Extension,” 61 pages. cited by applicant

Foreign Communication from Related Application—International Search Report and Written Opinion of the International Searching Authority, International Application No. PCT/US2023/025736, dated Oct. 17, 2023, 11 pages. cited by applicant

Foreign Communication from Related Application—International Search Report and Written Opinion of the International Searching Authority, International Application No. PCT/US2023/025717, dated Oct. 12, 2023, 12 pages. cited by applicant

First Office Action dated Apr. 15, 2024 (28 pages), U.S. Appl. No. 17/954,855, filed Sep. 28, 2022. cited by applicant

Filing Receipt, Specification and Drawings for U.S. Appl. No. 18/610,817, entitled “Electric Submersible Pump (ESP) Shroud System,” filed Mar. 20, 2024 49 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of and claims priority to U.S. patent application Ser. No. 17/369,526 filed Jul. 7, 2021, entitled “Electric Submersible Pump (ESP) Gas Slug Processor and Mitigation System,” which is hereby incorporated by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable.

REFERENCE TO A MICROFICHE APPENDIX

(2) Not applicable.

BACKGROUND

(3) Electric submersible pumps (hereafter “ESP” or “ESPs”) may be used to lift production fluid in

a wellbore. Specifically, ESPs may be used to pump the production fluid to the surface in wells with low reservoir pressure. ESPs may be of importance in wells having low bottomhole pressure or for use with production fluids having a low gas/oil ratio, a low bubble point, a high water cut, and/or a low API gravity. Moreover, ESPs may also be used in any production operation to increase the flow rate of the production fluid to a target flow rate.

(4) Generally, an ESP comprises an electric motor, a seal section, a pump intake, and one or more pumps (e.g., a centrifugal pump). These components may all be connected with a series of shafts. For example, the pump shaft may be coupled to the motor shaft through the intake and seal shafts. An electric power cable provides electric power to the electric motor from the surface. The electric motor supplies mechanical torque to the shafts, which provide mechanical power to the pump. Fluids, for example reservoir fluids, may enter the wellbore where they may flow past the outside of the motor to the pump intake. These fluids may then be produced by being pumped to the surface inside the production tubing via the pump, which discharges the reservoir fluids into the production tubing.

(5) The reservoir fluids that enter the ESP may sometimes comprise a gas fraction. These gases may flow upwards through the liquid portion of the reservoir fluid in the pump. The gases may even separate from the other fluids when the pump is in operation. If a large volume of gas enters the ESP, or if a sufficient volume of gas accumulates on the suction side of the ESP, the gas may interfere with ESP operation and potentially prevent the intake of the reservoir fluid. This phenomenon is sometimes referred to as a “gas lock” because the ESP may not be able to operate properly due to the accumulation of gas within the ESP.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 is an illustration of an electric submersible pump (ESP) assembly according to an embodiment of the disclosure.

(3) FIG. 2A, FIG. 2B, FIG. 2C, and FIG. 2D are an illustration of a gas separator assembly according to embodiments of the disclosure.

(4) FIG. 3 is an illustration of another gas separator assembly according to an embodiment of the disclosure.

(5) FIG. 4 is an illustration of an annulus in an interior of the gas separator assembly according to an embodiment of the disclosure.

(6) FIG. 5A is an illustration of an annular volume corresponding to the annulus in the interior of the gas separator assembly of FIG. 4.

(7) FIG. 5B is an illustration of a cross-sectional area of the annular volume of FIG. 5A.

(8) FIG. 6A is an illustration of another annulus and a spider bearing in an interior of the gas separator assembly according to an embodiment of the disclosure.

(9) FIG. 6B is an illustration of a cross-section of the spider bearing according to an embodiment of the disclosure.

(10) FIG. 6C is an illustration of yet another annulus and a plurality of spider bearings in an interior of the gas separator assembly according to an embodiment of the disclosure.

(11) FIG. 7A and FIG. 7B is a flow chart of a method according to an embodiment of the disclosure.

(12) FIG. 8A and FIG. 8B is a flow chart of another method according to an embodiment of the disclosure.

(13) FIG. 9 is an illustration of a tandem gas separator assembly according to an embodiment of the disclosure.

DETAILED DESCRIPTION

(14) It should be understood at the outset that although illustrative implementations of one or more embodiments are illustrated below, the disclosed systems and methods may be implemented using any number of techniques, whether currently known or not yet in existence. The disclosure should in no way be limited to the illustrative implementations, drawings, and techniques illustrated below, but may be modified within the scope of the appended claims along with their full scope of equivalents.

(15) As used herein, orientation terms “upstream,” “downstream,” “up,” “down,” “uphole,” and “downhole” are defined relative to the direction of flow of well fluid in the well casing. “Upstream” is directed counter to the direction of flow of well fluid, towards the source of well fluid (e.g., towards perforations in well casing through which hydrocarbons flow out of a subterranean formation and into the casing). “Downstream” is directed in the direction of flow of well fluid, away from the source of well fluid. “Down” and “downhole” are directed counter to the direction of flow of well fluid, towards the source of well fluid. “Up” and “uphole” are directed in the direction of flow of well fluid, away from the source of well fluid. “Fluidically coupled” means that two or more components have communicating internal passageways through which fluid, if present, can flow. A first component and a second component may be “fluidically coupled” via a third component located between the first component and the second component if the first component has internal passageway(s) that communicates with internal passageway(s) of the third component, and if the same internal passageway(s) of the third component communicates with internal passageway(s) of the second component.

(16) Gas entering a centrifugal pump of an electric submersible pump (ESP) assembly can cause various difficulties for a centrifugal pump. In an extreme case, the pump may become gas locked and become unable to pump fluid. In less extreme cases, the pump may experience harmful operating conditions when transiently passing a slug of gas. When in operation, the centrifugal pump rotates at a high rate of speed (e.g., about 3600 RPM) and relies on the continuous flow of reservoir liquid to both cool and lubricate its bearing surfaces. When this continuous flow of reservoir liquid is interrupted, even for a brief period of seconds, the bearings of the centrifugal pump may heat up rapidly and undergo significant wear, shortening the operational life of the centrifugal pump, thereby increasing operating costs due to more frequent change-out and/or repair of the centrifugal pump. Down time involved in repairing or replacing the centrifugal pump may also interrupt well production undesirably. In some operating environments, for example in some horizontal wellbores, gas slugs that persist for at least 10 seconds are repeatedly experienced. Some gas slugs may persist for as much as 30 seconds or more. The present disclosure teaches a new gas separator assembly that mitigates the effects of gas slugs.

(17) A gas separator assembly may comprise an inlet that feeds reservoir fluid to a first fluid mover (e.g., a multi-stage centrifugal pump or a rotating auger), and the first fluid mover drives reservoir fluid through to a second fluid mover, and the second fluid mover (e.g., a paddle wheel, a stationary auger, a vortex inducer) imparts a rotating motion to the reservoir fluid. The rotating reservoir fluid flows from the second fluid mover into a separation chamber. The rotation of the reservoir fluid in the separation chamber tends to separate gas phase fluid from liquid phase fluid. Due to the rotation of the reservoir fluid, the relatively lower density gas phase fluid tends to concentrate near a centerline axis of the gas separator assembly (e.g., near a drive shaft of the gas separator assembly), and the relatively higher density liquid phase fluid tends to concentrate near an inside wall of a housing or separation chamber of the gas separator assembly. The fluid near the centerline axis enters a gas phase discharge of the gas separator assembly and exits the gas separator assembly to an annulus formed between the wellbore and the outside of the ESP assembly; the fluid near the inside wall enters a liquid phase discharge of the gas separator

assembly and is directed downstream to another stage of the gas separator assembly or to an inlet of the centrifugal pump assembly. In this way the reservoir fluid that is fed downstream to the inlet of the centrifugal pump assembly may be said to be a liquid enriched reservoir fluid or a liquid enriched fraction of the reservoir fluid.

(18) In the circumstance of a large slug of gas reaching the ESP assembly, however, a conventional gas separator assembly may quickly fill with gas. In this circumstance, there is no liquid phase fraction to separate and forward on as a liquid enriched fraction of the reservoir fluid. When the large slug of gas first reaches the gas separator assembly, for a short period of time, liquid phase fluid retained within the passageways of the fluid mover may mix with the gas of the slug, and a blend of gas phase fluid and liquid phase fluid may be supplied briefly by the gas separator assembly to the centrifugal pump assembly. This blend of gas phase fluid and liquid phase fluid may provide lubrication to bearing surfaces of the centrifugal pump assembly, provide heat transfer away from bearing surfaces of the centrifugal pump assembly, and avoid putting the centrifugal pump assembly into a gas lock condition. But the passageways of the fluid mover (e.g., vane channels of impellers and diffusers) are limited in volume, and the retained liquid phase fluid is rapidly depleted in the presence of a large gas slug. The present disclosure teaches creating additional internal volumes inside the gas separator assembly between fluid mover stages (e.g., between centrifugal pump stages) that operate as reservoirs of liquid phase fluid that can extend the transition time from normal operation to a condition in which the gas separator assembly is completely gas filled and supplies no liquid phase fluid to the inlet of the centrifugal pump assembly.

(19) Turning now to FIG. 1 a well site environment **100**, according to one or more aspects of the disclosure, is described. The well site environment **100** comprises a wellbore **102** that is at least partially cased with casing **104**. As depicted in FIG. 1, the wellbore **102** has a deviated or horizontal portion **106**, but the electric submersible pump (ESP) assembly **132** described herein may be used in a wellbore **102** that does not have a deviated or horizontal portion **106**. The well site environment **100** may be at an on-shore location or at an off-shore location. The ESP assembly **132** in an embodiment comprises a sensor package **120**, an electric motor **122**, a seal unit **124**, a gas separator assembly **126**, and a centrifugal pump assembly **128**. The centrifugal pump assembly may couple to a production tubing **134** via a connector **130**. An electric cable **135** may attach to the electric motor **122** and extend to the surface **158** to connect to an electric power source. The gas separator assembly **126** comprises inlet ports **136** and gas phase discharge ports **138**. The casing **104** and/or wellbore **102** may have perforations **140** that allow reservoir fluid **142** to pass from the subterranean formation through the perforations **140** and into the wellbore **102**. In an embodiment, a distance between the inlet ports **136** and the gas discharge ports **138** are less than 500 feet and at least 4 feet, at least 6 feet, at least 8 feet, at least 10 feet, at least 12 feet, at least 14 feet, at least 16 feet, at least 18 feet, at least 20 feet, at least 22 feet, at least 24 feet, at least 26 feet, at least 28 feet, at least 30 feet, at least 32 feet, at least 35 feet, at least 40 feet, at least 45 feet, at least 50 feet, at least 60 feet, at least 70 feet, at least 80 feet, at least 90 feet, at least 100 feet, at least 120 feet, or at least 140 feet.

(20) The reservoir fluid **142** may flow uphole towards the ESP assembly **132** and into the inlet ports **136**. The reservoir fluid **142** may comprise a liquid phase fluid. The reservoir fluid **142** may comprise a gas phase fluid mixed with a liquid phase fluid. The reservoir fluid **142** may comprise only a gas phase fluid (e.g., simply gas). Over time, the gas to fluid ratio of the reservoir fluid **142** may change dramatically. For example, in the horizontal portion **106** of the wellbore gas may build up in high points in the roof of the wellbore **102** and after accumulating sufficiently may “burp” out of these high points and flow downstream to the ESP assembly **132** as what is commonly referred to as a gas slug. Thus, immediately before a gas slug arrives at the ESP assembly **132**, the gas fluid ratio of the reservoir fluid **142** may be very low (e.g., the reservoir fluid **142** at the ESP assembly **132** is mostly liquid phase fluid); when the gas slug arrives at the ESP assembly **132**, the gas fluid

ratio is very high (e.g., the reservoir fluid **142** at the ESP assembly **132** is entirely or almost entirely gas phase fluid); and after the gas slug has passed the ESP assembly **132**, the gas fluid ratio may again be very low (e.g., the reservoir fluid **142** at the ESP assembly **132** is mostly liquid phase fluid).

(21) Under normal operating conditions (e.g., reservoir fluid **142** is flowing out of the perforations **140**, the ESP assembly **132** is energized by electric power, the electric motor **122** is turning, and a gas slug is not present at the ESP assembly **132**), the reservoir fluid **142** enters the inlets **136**, the reservoir fluid **142** is separated by the gas separator assembly **138** into a gas phase fluid (or a mixed-phase fluid having a higher gas liquid ratio than the reservoir fluid **142** entering the inlet ports **136**) and a liquid phase fluid (or a mixed-phase fluid having a lower gas liquid ratio than the reservoir fluid **142** entering the inlet ports **136**). The gas phase fluid is discharged via the gas phase discharge ports **138**, and the liquid phase fluid is flowed downstream to the centrifugal pump assembly **128** as liquid phase fluid **154**. Under normal operating conditions, the gas phase fluid that is discharged into the annulus between the casing **104** and the outside of the ESP assembly **132** may comprise both gas phase fluid **150** that rises uphole in the wellbore **102** and liquid phase fluid **152** that falls downhole in the wellbore **102**. The centrifugal pump assembly **128** flows the liquid phase fluid **154** (e.g., a portion of the reservoir fluid **142**) up the production tubing **134** to a wellhead **156** at the surface **158**.

(22) An orientation of the wellbore **102** and the ESP assembly **132** is illustrated in FIG. **1** by an x-axis **160**, a y-axis **162**, and a z-axis **164**. In an embodiment, the centrifugal pump assembly **128** comprises one or more centrifugal pump stages, where each stage comprises an impeller that is mechanically coupled to a drive shaft within the centrifugal pump assembly **128** and a corresponding diffuser that is stationary and retained by a housing of the centrifugal pump assembly **128**. In an embodiment, the impellers may comprise a keyway that mates with a corresponding keyway on the drive shaft of the centrifugal pump assembly **128** and a key may be installed into the two keyways, wherein the impeller may be mechanically coupled to the drive shaft of the centrifugal pump assembly.

(23) Turning now to FIG. **2A**, FIG. **2B**, FIG. **2C**, and FIG. **2D**, further details of the gas separator assembly **126** are described. The gas separator assembly **126** comprises a base **403**, a housing **312**, a crossover **350**, and a head **355**. The base **410** has the inlet ports **136** and couples threadingly at a downstream end with an upstream end of the housing **312**, for example via threaded coupling **403**. In some contexts, the base **410** may be said to be mechanically coupled to the housing **312**. In an embodiment, the base **410** couples to the seal unit **124**, for example with a bolted connection (not shown) or a threaded coupling. The housing **312** may be a cylindrical hollow metal pipe. In an embodiment, an inside of the housing **312** may be machined or drilled at one or more locations to create slots or shallow holes for fixing and retaining components within the housing **312**, for example diffusers or other components.

(24) In an embodiment, the housing **312** encloses a plurality of centrifugal pump stages **405**, for example a first centrifugal pump stage **405A** and a second centrifugal pump stage **405B**. Each centrifugal pump stage **405** comprises an impeller **406** mechanically coupled to a drive shaft **172** of the gas separator assembly **126** and a diffuser **408** that is retained and held stationary by the housing **312**. In an embodiment, the impeller **406** may have a keyway that mates with a keyway in the drive shaft **172** and the keyway of the impeller **406** may be secured to the keyway in the drive shaft **172** by a key. In an embodiment, the impeller **406** may be mechanically coupled to the drive shaft **172** in a different way. When the drive shaft **172** turns, the impeller **406** turns. The first centrifugal pump stage **405A** comprises a first impeller **406A** and a first diffuser **408A**; the second centrifugal pump stage **405B** comprises a second impeller **406B** and a second diffuser **408B**. While two centrifugal pump stages **405A** and **405B** are illustrated in FIG. **2A**, FIG. **2B**, and FIG. **2C**, in another embodiment, there may be a single centrifugal pump stage **405**, three centrifugal pump stages **405**, four centrifugal pump stages **405**, five centrifugal pump stages **405**, six centrifugal

pump stages **405**, or more centrifugal pump stages **405** located between the base **410** and the fluid reservoir **170**. The centrifugal pump stages **405** may be referred to as a first fluid mover in some contexts. In an embodiment, the centrifugal pump stages **405** of the gas separator assembly **126** are replaced by another fluid mover mechanism, for example replaced by an auger mechanically coupled to the drive shaft **172** (as illustrated in FIG. 2C and FIG. 2D), one or more impeller mechanically coupled to the drive shaft **172** (e.g., without a corresponding diffuser), and/or a paddle wheel mechanically coupled to the drive shaft **172**. The auger mechanically coupled to the drive shaft **172** may be referred to as a rotating auger **1302** and may comprise one or more vanes **1324** and a shaft **1318** that encloses the drive shaft **172**. The rotating auger **1302** may define a keyway **1376** in its inside that may be aligned with a keyway **1374** defined in an outside of the drive shaft **172**. A key **1378** may be inserted into the keyways **1374**, **1376** to mechanically couple the rotating auger **1302** to the drive shaft **172**. As shown in FIG. 2D, the rotating auger **1302** may have a first vane **1324a** and a second vane **1324b**.

(25) In an embodiment, the drive shaft **172** is mechanically coupled to a drive shaft of the seal unit **124**, and the drive shaft of the seal unit **124** is mechanically coupled to a drive shaft of the electric motor **122**. Thus, the drive shaft **172** and the impellers **406** (e.g., impellers **406A** and **406B** in FIG. 2A, FIG. 2B, and FIG. 2C) of the one or more centrifugal pump stages **405** are turned indirectly by the electric motor **122** when it is energized by electric power via the electric cable **135**. The drive shaft **172** is mechanically coupled to a drive shaft of the centrifugal pump assembly **128** and transfers rotational power to the drive shaft of the centrifugal pump assembly **136** and to impellers of the centrifugal pump stages of the centrifugal pump assembly **136**. The several different drive shaft mechanical couplings may be provided by splines cut in the mating ends of shafts and coupled by a spline coupler or hub. In another embodiment, the drive shaft mechanical couplings may be provided by other devices.

(26) The housing **312** also encloses a fluid reservoir **170**. In an embodiment, the fluid reservoir **170** is formed as an annulus between the outside of the drive shaft **172** and the interior wall of the housing **312**. In an embodiment (as illustrated in FIG. 2 B), the fluid reservoir **170** is formed by a sleeve retained within the housing **312** that has an inlet at an upstream end of the fluid reservoir **170** that is fluidically coupled to an outlet of the second diffuser **408A** and has an outlet **304** at a downstream end of the fluid reservoir **170** that is fluidically coupled to an upstream end of a second fluid mover, for example a stationary auger **302**. The fluid reservoir **170** may retain mostly liquid phase fluid when the ESP assembly **132** is experiencing normal operating conditions (e.g., when the electric motor **122** is energized and turning, when reservoir fluid **142** is entering the wellbore **102** and flowing in the inlet ports **136**, and in the absence of a gas slug), and this liquid phase fluid can be mixed progressively with gas when the ESP assembly **132** receives a gas slug to extend the time that the gas separator assembly **126** is able to continue to supply at least some liquid phase fluid to the centrifugal pump assembly **128**.

(27) For example, at a first point in time, before the gas slug arrives at the inlet ports **136**, the outlet **304** of the fluid reservoir **170** may provide fluid having a first gas liquid ratio (GLR) to the stationary auger **302**. As gas from the gas slug enters the inlet ports **136**, at a second point in time (after the first point in time) the gas mixes with the fluid in the fluid reservoir **170**, and the outlet **304** of the fluid reservoir **170** may provide fluid having a second GLR to the stationary auger **302**, where the second GLR is greater than the first GLR. At a third point in time (after the second point in time) the gas continues to mix with the fluid in the fluid reservoir **170**, and the outlet **304** of the fluid reservoir **170** may provide fluid having a third GLR to the stationary auger **302**, where the third GLR is greater than the second GLR. At a fourth point in time (after the third point in time), when the gas slug passes the ESP assembly **132** and is no longer be drawn into the inlet ports **136**, the reservoir fluid **142** entering the inlet ports **136** may again be primarily liquid phase fluid, and the outlet **304** of the fluid reservoir **170** may provide fluid having a fourth GLR to the stationary auger **302**, where the fourth GLR is less than the third GLR. At a fifth point in time (after the fourth

point in time), the outlet **304** of the fluid reservoir **170** may provide fluid having a fifth GLR to the stationary auger **302**, where the fifth GLR is less than the fourth GLR and approximately equal to the first GLR. It is noted that without the primarily liquid phase fluid retained in the fluid reservoir **170** at the time that the gas slug arrived at the ESP assembly **132** and the inlet ports **136**, the GLR would have risen very quickly and would have flowed gas unmixed from the outlets of the second diffuser **408B** to the auger **302**, from the stationary auger **302** to the separation chamber **303**, from the separation chamber **303** to the liquid phase discharge **316** of the crossover **350**, and from the liquid phase discharge **316** to the inlet of the centrifugal pump assembly **128**, with the undesirable effect that the bearings of the centrifugal pump assembly **128** would lose lubrication, would rapidly heat up, would rapidly degrade, and likely would leave the centrifugal pump stages in the centrifugal pump assembly **128** in a gas lock situation. In an embodiment, the gas separator assembly **126** may also have one or more centrifugal pump stages between the fluid reservoir **170** and the stationary auger **302**.

(28) The housing **312** also encloses a stationary auger **302**. In one or more embodiments, the stationary auger **302** is disposed or positioned within a sleeve **322**. The centrifugal pump stages **405** communicates or forces reservoir fluid **142** received at the one or more inlet ports **136** through the fluid reservoir **170** and through the stationary auger **302**. In an embodiment, an outside edge of the stationary auger **302** engages sealingly with an inside surface **330** of the sleeve **322**, and the flow of reservoir fluid **142** through the sleeve **322** is hence confined to the passageway or passageways defined by the stationary auger **302**. The sleeve **322** may be disposed or positioned within and retained by the housing **312**. In an embodiment, the stationary auger **302** and the sleeve **322** may be built or manufactured as a single component.

(29) In an embodiment, there is no sleeve **322** and the stationary auger **302** is disposed within the inside of the housing **312**. The stationary auger **202** may be retained by the inside of the housing **312**. In an embodiment, the stationary auger **302** engages sealingly with an inside surface of the housing **312**. In an embodiment, there is a space between the outside edges of the stationary auger **302** and the inside surface **330** of the sleeve **332** or a space between the outside edges of the stationary auger **302** and the inside surface of the housing **312**.

(30) In one or more embodiments, the stationary auger **302** comprises one or more helixes or vanes **324**. In one or more embodiments, the helixes or vanes **324** may be crescent-shaped. In one or more embodiments, the stationary auger **302** comprises one or more helixes or vanes **324** disposed about a solid core, for example shaft **318** that encloses the drive shaft **172**, or an open core (for example, a coreless auger or an auger flighting). The stationary auger **302** may cause the reservoir fluid **142** to be separated into a liquid phase fluid **428** and gas phase fluid **426** based, at least in part, on rotational flow of the reservoir fluid **142**.

(31) For example, the one or more helixes or vanes **324** may impart rotation to the reservoir fluid **142** as the reservoir fluid **142** flows through, across or about the one or more helixes or vanes **324**. The stationary auger **302**, then, can be referred to as a fluid mover at least because it imparts a rotating motion to the reservoir fluid **142** as the reservoir fluid **142** flows through the stationary auger **302**. For example, fluid mover **310** forces the reservoir fluid **142** at a velocity or flow rate into the sleeve **322** and up or across the one or more helixes or vanes **324** of stationary auger **302**. The rotation of the reservoir fluid **142** induced by the stationary auger **302** may be based, at least in part, on the velocity or flow rate of the reservoir fluid **142** generated by the centrifugal pump stages **405**. For example, the centrifugal pump stages **405** may increase the flow rate or velocity of the reservoir fluid **142** to increase rotation of the reservoir fluid **142** through the stationary auger **302** to create a more efficient and effective separation of the reservoir fluid **142** into a plurality of phases, for example, a liquid phase fluid **428** and a gas phase fluid **426**. As the reservoir fluid **142** flows through the stationary auger **302**, centrifugal forces, static friction or both, cause the heavier component of the reservoir fluid **142**, a liquid phase fluid **428**, to circulate along an outer perimeter of the stationary auger **112** while the lighter component of the reservoir fluid **142**, the gas phase

fluid **426**, is circulated along an inner perimeter of the stationary auger **302**. In one or more embodiments, reservoir fluid **142** may begin to separate while flowing through stationary auger **302**. In one or more embodiments, the liquid phase fluid **428** may comprise residual gas that did not separate into the gas phase fluid **426**. However, the embodiments discussed herein reduce this residual gas to protect the centrifugal pump assembly **128** from gas build-up or gas lock.

(32) In an embodiment, the stationary auger **302** is not present and instead a different kind of second fluid mover is provided. The second fluid mover may be provided by an auger mechanically coupled to the drive shaft **172**, a paddle wheel mechanically coupled to the drive shaft **172**, a centrifuge rotor mechanically coupled to the drive shaft **172**, or an impeller mechanically coupled to the drive shaft **172** that induce rotating motion of the reservoir fluid **142**. In an embodiment, a third fluid mover is provided downstream of the stationary auger **302**, for example a paddle wheel may be installed downstream of the stationary auger **172** that induces and/or increases rotating motion of the reservoir fluid **142**.

(33) A separation chamber **303** is provided downstream of the second fluid mover (e.g., the stationary auger **302**) and downstream of the optional third fluid mover. An upstream end of the separation chamber **303** is fluidically coupled to a downstream end or an outlet of the stationary auger **302** or other second fluid mover. Alternatively, the upstream end of the separation chamber **303** is fluidically coupled to a downstream end or an outlet of the optional third fluid mover and is fluidically coupled to the third fluid mover and, via the third fluid mover, fluidically coupled to the second fluid mover. The separation chamber **303** is defined by an annulus formed between the inside of the housing **312** and the outside of the drive shaft **172**. In an embodiment, the separation chamber **303** is less than 36 inches long and at least 4 inches long, at least 6 inches long, at least 8 inches long, at least 10 inches long, at least 12 inches long, or at least 14 inches long. In an embodiment, the separation chamber is at least 6 inches long and less than 17 inches long. The stationary auger **302** (or other second fluid mover and/or third fluid mover) induces a rotating motion in the reservoir fluid **142**. As the reservoir fluid **142** exits the stationary auger **302** (or other second fluid mover and/or third fluid mover) and enters the separation chamber **303**, this rotating motion of the reservoir fluid **142** continues. The rotating motion of the reservoir fluid **142** within the separation chamber **303** induces gas phase fluid (which is less dense than the liquid phase fluid) to concentrate near the drive shaft **172** and the liquid phase fluid to concentrate near the inside surface of the housing **312**.

(34) In one or more embodiments, the separated fluids (for example, liquid phase fluid **428** and gas phase fluid **426**) are directed to a crossover **350**. For example, the crossover **350** may be disposed or positioned at a downstream end of the separation chamber **303** or housing **312**. In some contexts, the crossover **350** may be referred to as a gas flow path and liquid flow path separator. The crossover **350** may comprise a plurality of channels or define a plurality of channels, for example, a gas phase discharge **314** (a first pathway) and a liquid phase discharge **316** (a second pathway). A gas phase fluid **426** of the reservoir fluid **142** may be discharged through the gas phase discharge **314**, out the gas phase discharge ports **138**, and a liquid phase fluid **428** of the reservoir fluid **142** may be discharged through the liquid phase discharge **316**. In one or more embodiments, gas phase discharge **314** may correspond to any one or more discharge ports **138** of FIG. **1**. In one or more embodiments, any one or more of the gas phase discharge ports **314** and the one or more liquid phase discharge ports **316** may be defined by a channel or pathway having an opening, for example, a teardrop shaped opening, a round opening, an elliptical opening, a triangular opening, a square opening, or another shaped opening. The crossover **350** may be threadingly coupled at an upstream end by threaded coupling **351** to a downstream end of the housing **312**. The crossover **350** may be threadingly coupled at a downstream end by threaded coupling **357** to a head **355**. Alternatively, the head **355** may be integrated with the head **355** rather than threadingly coupled to the head **355**. The head **355** may provide bolt holes for coupling to an upstream end of the centrifugal pump assembly **128**. In some contexts, the crossover **350** may be said to be

mechanically coupled at an upstream end to a downstream end of the housing **312**. When the crossover **350** and the head **355** are not integrated as a single component, the crossover **350** may be said to be mechanically coupled at a downstream end to an upstream end of the head **355**.

(35) Turning now to FIG. **3**, another embodiment of the gas separator assembly **126** is described. In an embodiment, the gas separator assembly **126** of FIG. **3** may be similar to the gas separator assembly **126** of FIG. **2A**, FIG. **2B**, FIG. **2C**, and FIG. **2D** but in addition may comprise a plurality of fluid reservoirs. In an embodiment, the gas separator assembly **126** may comprise a plurality of fluid reservoirs separated by a plurality of centrifugal pump stages **405**, **415**, **425**.

(36) For example, a second set of centrifugal pump stages **415** may be located within the housing **312** downstream of the fluid reservoir **170** and upstream of a second fluid reservoir **174**. The second set of centrifugal pump stages **415** comprise a third pump stage **415A** that comprises third impeller **416A** mechanically coupled to the drive shaft **172** and a third diffuser **418A** held stationary and retained by the housing **312**, and a fourth pump stage **415B** that comprises a fourth impeller **416B** mechanically coupled to the drive shaft **172** and a fourth diffuser **418B** held stationary and retained by the housing **312**. One or more inlets of the third impeller **416A** are fluidically coupled to the fluid reservoir **170**. In another embodiment, the second set of centrifugal pump stages **415** may comprise a single centrifugal pump stage, three centrifugal pump stages, four centrifugal pump stages, five centrifugal pump stages, six centrifugal pump stages, or some other number of centrifugal pump stages. When the drive shaft **172** turns, the third impeller **416A** and the fourth impeller **416B** turn.

(37) In an embodiment, the second fluid reservoir **174** is formed as an annulus between the outside of the drive shaft **172** and the interior wall of the housing **312**. Alternatively, the second fluid reservoir **174** is formed as an annulus between the outside of the drive shaft **172** and an inside of a sleeve retained within the housing **312** that has an inlet at an upstream end of the second fluid reservoir **174** fluidically coupled to the outlets of the fourth diffuser **418B** of the fourth centrifugal pump stage **415B** and an outlet at a downstream end of the second fluid reservoir **174** that is fluidically coupled to another set of centrifugal pump stages, for example centrifugal pump stages **425**.

(38) A third fluid reservoir **176** may be located downstream of the second fluid reservoir **174** and upstream of a third set of centrifugal pump stages **425**. In an embodiment, the third fluid reservoir **176** is formed as an annulus between the outside of the drive shaft **172** and the interior wall of the housing **312**. Alternatively, the third fluid reservoir **176** is formed as an annulus between the outside of the drive shaft **172** and an inside of a sleeve retained within the housing **312** that has an inlet at an upstream end of the third fluid reservoir **176** and an outlet at a downstream end of the third fluid reservoir **176**. Additional centrifugal pump stages (not shown) may be located between the second fluid reservoir **174** and the third fluid reservoir **176**, for example between the cut lines in FIG. **3**. Additional fluid reservoirs (not shown) may be located between the second fluid reservoir **174** and the third fluid reservoir **176**, for example between the cut lines in FIG. **3**. In an embodiment, one or more of the centrifugal pumps **405**, **415**, **425** may be provided by a different type of fluid mover, for example an auger mechanically coupled to the drive shaft **172**, a paddle wheel mechanically coupled to the drive shaft **172**, or an impeller mechanically coupled to the drive shaft **172**. In an embodiment, an outlet of the second fluid reservoir **174** is fluidically coupled to an inlet of the third fluid reservoir **176**. In an embodiment, the second fluid reservoir **174** may be fluidically coupled to the inlet of the third fluid reservoir **176** via internal passageways of one or more centrifugal pump stages located between the second and third fluid reservoirs **174**, **176**.

(39) In an embodiment, the second fluid reservoir **174** and the third fluid reservoir **176** may not be separated by any centrifugal pump stages but may feature a spider bearing fixed to and retained by the inside of the housing **312** to support the drive shaft **170**. A large fluid reservoir may be formed by stringing a plurality of fluid reservoirs together with spider bearings in between to support the drive shaft **170** at regular intervals, for example every 6 inches, every 8 inches, every 9 inches,

every 10 inches, every 11 inches every 12 inches, every 13 inches, every 14 inches, or every 16 inches. The spacing between spider bearings may be dependent on a diameter of the drive shaft **170**. For example, if the drive shaft **170** has a smaller diameter, the spider bearings may be placed more closely together; if the drive shaft **170** has a larger diameter, the spider bearings may be placed further apart.

(40) The second fluid reservoir **174** provides the same function as the fluid reservoir **170** and extends yet further the amount of time that the ESP assembly **132** can sustain a gas slug (e.g., a bigger gas slug, a more extensive gas slug) without losing liquid phase fluid flow **154** to the centrifugal pump assembly **128**, without bearings in the centrifugal pump assembly **128** overheating, and without the centrifugal pump assembly **128** experiencing gas lock. The third fluid reservoir **176** (and possibly additional fluid reservoirs between the second fluid reservoir **174** and the third fluid reservoir **176**) again provides greater ability to sustain a gas slug for a longer period of time without losing liquid phase fluid flow **154** to the centrifugal pump assembly **128**. The greater the sum of the volume of the fluid reservoir **170**, the volume of the second fluid reservoir **174**, and the volume of the third fluid reservoir (and the volumes of any further intervening fluid reservoirs), the longer duration of gas slug (the bigger the gas slug) that the ESP assembly **132** can sustain.

(41) In an embodiment, the gas separator assembly **126** has one or more centrifugal pump stages downstream of the third fluid reservoir **176** and upstream of a paddle wheel **327** (In FIG. 3 the stationary auger **302** is replaced with a paddle wheel **327** that imparts rotating motion to the reservoir fluid **142** before it flows into the separation chamber **303**), for example a fifth centrifugal pump stage **425A** and a sixth centrifugal pump stage **425B**. The fifth centrifugal pump stage **425A** comprises a fifth impeller **426A** mechanically coupled to the drive shaft **172** and a fifth diffuser **428A** retained and held stationary by the housing **312**. The sixth centrifugal pump stage **425B** comprises a sixth impeller **426B** mechanically coupled to the drive shaft **172** and a sixth diffuser **428B** retained and held stationary by the housing **312**. When the drive shaft **172** turns, the fifth impeller **426A** and the sixth impeller **426B** are turned. While two centrifugal pump stages **425A**, **425B** are illustrated downstream of the third fluid reservoir **176** and upstream of the paddle wheel **303**, in another embodiment a single centrifugal pump stage, three centrifugal pump stages, four centrifugal pump stages, five centrifugal pump stages, six centrifugal pump stages, or more centrifugal pump stages may be located downstream of the third fluid reservoir **176** and upstream of the paddle wheel **327** in the gas separator assembly **126**. The paddle wheel **327** is mechanically coupled to the drive shaft **172**.

(42) Turning now to FIG. 4, the fluid reservoir **170** is illustrated as an annulus defined between the drive shaft **172** and the inner surface **171** (e.g., the inner wall of the housing **312** or a sleeve inside the inner wall of the housing **312**). The annular volume of the annulus defined by the fluid reservoir is shown better in FIG. 5A and FIG. 5B. The volume may be found as the cross-sectional area of the annular volume **180** (best seen in FIG. 5B) multiplied by the length of the fluid reservoir **170** indicated as 'L1' in FIG. 4 and in FIG. 5A. The cross-sectional area of the annular volume **180** can be found as the difference of the area of a circle of diameter D2 (the inside diameter of the housing **312** or the inside diameter of the sleeve) and the area of a circle of diameter D1 (the diameter of the drive shaft **172**). By increasing the sum volume of fluid reservoirs inside the gas separator assembly **126**, the gas separator assembly **126** is able to sustain gas slugs of increasing duration.

(43) In an embodiment, the fluid reservoir **170** is at least 2 inches long and less than 14 inches long. In an embodiment, the fluid reservoir **170** is at least 6 inches long and less than 14 inches long. In an embodiment, the fluid reservoir **170** is at least 14 inches long and less than 28 inches long. In an embodiment, the fluid reservoir **170** is at least 17 inches long and less than 34 inches long. In an embodiment, the fluid reservoir **170** is at least 24 inches long and less than 42 inches long. In an embodiment, the annular volume **180** of the fluid reservoir **170** is at least 18 cubic inches and less than 1000 cubic inches. In an embodiment, the annular volume **180** of the fluid reservoir **170** is at

least 50 cubic inches and less than 1000 cubic inches. In an embodiment, the fluid reservoir **170** may comprise one or more spider bearings to support the drive shaft **172** as discussed further hereinafter.

(44) In an embodiment, the gas separator assembly **126** may be less than 500 feet long and at least, 5 feet long, at least 8 feet long, at least 10 feet long, at least 12 feet long, at least 14 feet long, at least 16 feet long, at least 18 feet long, at least 20 feet long, at least 22 feet long, at least 24 feet long, at least 26 feet long, at least 28 feet long, at least 30 feet long, at least 32 feet long, at least 34 feet long, at least 40 feet long, at least 50 feet long, at least 60 feet long, at least 70 feet long, at least 80 feet long, at least 90 feet long, at least 100 feet long, at least 120 feet long, or at least 140 feet long. With long gas separator assemblies **126**, the gas separator assembly may comprise a first housing that threadingly couples with a second housing, and the first housing and second housing joined together contain the centrifugal pump stages, the fluid reservoirs, and the stationary auger **302** of the gas separator assembly **126**. With long gas separator assemblies **126**, the drive shaft **172** may comprise two drive shafts that are coupled together by a spline coupling.

(45) In an embodiment, during normal operation (e.g., there is no gas slug present at the inlet ports **136**), liquid phase fluid may fill the annulus **210** from the downhole end of the gas separator assembly **126** (e.g., at the fluid inlets **136**) to the level of the discharge ports **138**. This liquid phase fluid may also mix with gas at the inlet ports **136** and in the centrifugal pump stages **405** when a gas slug hits the ESP assembly **132**. Thus, the longer the gas separator assembly **126**, the larger the volume of liquid phase fluid retained in the annulus **210** and the longer the ESP assembly **132** can sustain a gas slug while still feeding some liquid phase fluid to the centrifugal pump assembly **128**. Thus, extending the length of the gas separator assembly **126** with fluid reservoirs **170**, **174**, **176** also may create additional liquid fluid reserves in the annulus **210**.

(46) Turning now to FIG. 6A, an annular volume **182** is illustrated. A spider bearing **184** is illustrated in about a middle of the length L2 of the annular volume **182**. By supporting the drive shaft **172** in a middle portion, the length L2 can be made greater, for example can be increased to 16 inches, 18 inches, 20 inches, 22 inches, 24 inches, 26 inches, or 28 inches. The use of spider bearings **184** can readily increase the sum of volumes of fluid reservoir within the gas separator and pump assembly **126**. In FIG. 6B a different view of the spider bearing **184** is illustrated. The spider bearing **184** may comprise three struts **188** that stabilize a central bearing **186** of the spider bearing **184**. The struts **188** may be secured by the housing **312**. The struts **188** may take a shape of vanes oriented so as to minimally block the communication of reservoir fluid **142** through the spider bearing **184**, between the struts **188**. The spider bearing **184** provides fluid communication paths between the struts **188**. While FIG. 6A and FIG. 6B illustrate a spider bearing **184** with three struts **188**, the spider bearings **184** may comprise two struts, four struts, five struts, or some greater number of struts **188**. In FIG. 6C, the number of spider bearings **184** may be increased to any number, thereby increasing the volume annular volume defined by the fluid reservoir **170**, **174**, **176**. As shown in FIG. 6C, three spider bearings **184a**, **184b**, **184c** are used and may provide a length L3 of the fluid reservoir **170**, **174**, **176** of 24 inches, 32, inches, 40 inches, 44 inches, 48 inches, 52 inches, or 56 inches.

(47) In an embodiment, the drive shaft **172** has an outside diameter of about $\frac{7}{8}$ inches (e.g., about 0.875 inches), and the gas separator assembly **126** has an outside diameter of about 4 inches. In this case, the inside diameter of the housing **312** or of the sleeve inside the inner wall of the housing **312** is about $3\frac{1}{2}$ inches (e.g., 3.5 inches). These dimensions give a D1 value of about 0.875 inches, a D2 value of about 3.5 inches. The area of the cross-section in FIG. 5B for these values of D1 and D2 can be calculated to be about 9.0198 square inches. A corresponding annular volume can be calculated for a plurality of different values for L1 as per below:

(48) TABLE-US-00001 Value of L1 Corresponding annular volume

2"	18.040 cubic inches
4"	36.079 cubic inches
6"	54.119 cubic inches
8"	72.158 cubic inches
10"	90.198 cubic inches
12"	108.24 cubic inches
14"	126.28 cubic inches

(49) In an embodiment, the drive shaft **172** has an outside diameter of about 11/16 inches (e.g., about 0.6875 inches), and the gas separator assembly **126** has an outside diameter of about 4 inches. In this case, the inside diameter of the housing **312** or of the sleeve inside the inner wall of the housing **312** is about 3½ inches (e.g., 3.5 inches). The area of the cross-section in FIG. 5B for these values of D1 and D2 can be calculated to be about 9.2499 square inches. A corresponding annular volume can be calculated for a plurality of different values for L1 as per below:

(50) TABLE-US-00002 Value of L1 Corresponding annular volume 2" 18.500 cubic inches 4" 37.000 cubic inches 6" 55.499 cubic inches 8" 73.999 cubic inches 10" 92.499 cubic inches 12" 111.00 cubic inches 14" 129.50 cubic inches

(51) In an embodiment, the drive shaft **172** has an outside diameter of about 1 3/16 inches (e.g., about 1.1875 inches), and the gas separator assembly **126** has an outside diameter of about 5.38 inches. In this case, the inside diameter of the housing **312** or of the sleeve inside the inner wall of the housing **312** is about 4.77 inches. The area of the cross-section in FIG. 5B for these values of D1 and D2 can be calculated to be about 16.763 square inches. A corresponding annular volume can be calculated for a plurality of different values for L1 as per below:

(52) TABLE-US-00003 Value of L1 Corresponding annular volume 2" 33.526 cubic inches 4" 67.052 cubic inches 6" 100.58 cubic inches 8" 134.10 cubic inches 10" 167.63 cubic inches 12" 201.16 cubic inches 14" 234.68 cubic inches

(53) In an embodiment, the drive shaft **172** has an outside diameter of about 1 inch, and the gas separator assembly **126** has an outside diameter of about 5.38 inches. In this case, the inside diameter of the housing **312** or of the sleeve inside the inner wall of the housing **312** is about 4.77 inches. The area of the cross-section in FIG. 5B for these values of D1 and D2 can be calculated to be about 17.085 square inches. A corresponding annular volume can be calculated for a plurality of different values for L1 as per below:

(54) TABLE-US-00004 Value of L1 Corresponding annular volume 2" 34.170 cubic inches 4" 68.340 cubic inches 6" 102.51 cubic inches 8" 136.68 cubic inches 10" 170.85 cubic inches 12" 205.02 cubic inches 14" 239.19 cubic inches

(55) The diameter of the drive shaft **172** and the inside diameter of the housing **312** or sleeve may be determined by the wellbore environment the ESP assembly **132** may be deployed to. By varying the length L1, however, more or less annular volume may be created in the fluid reservoir **170**. More annular volume provides further buffer or reserve against gas slugs. At the same time, the length L1 may not be increased indefinitely because the drive shaft **172** may be unsupported and unstabilized in the fluid reservoir **170**. In an embodiment, this length L1 may desirably be restricted to less than 16 inches, less than 15 inches, less than 14 inches, less than 13 inches, less than 12 inches, less than 11 inches, or less than 10 inches. The maximum prudent length of L1 depends upon the diameter of the drive shaft **172**—the value of D1. A greater diameter drive shaft **172** may allow a relatively larger maximum length of L1 while a smaller diameter drive shaft **172** may allow a relatively smaller maximum length of L1. Greater annular volume—and hence greater ability to sustain gas slugs of long duration—can be provided either by increasing the length L1 or by increasing the number of fluid reservoirs within the gas separator assembly **126**. Greater annular volume can be provided by increasing the length L1 by adding spider bearings **184** and desirable intervals within a single fluid reservoir to maintain the desired stability and support for the drive shaft **172**.

(56) It is noted that the substantial open volumes between centrifugal pump stages and a stationary auger taught herein are not conventionally included in gas separator assemblies because additional materials are required to do this (longer housing **312**, for example), and longer spans where the drive shaft **172** is not supported occur.

(57) Turning now to FIG. 7A and FIG. 7B, a method **900** is described. In an embodiment, the method **900** is a method of lifting liquid in a wellbore. At block **902**, the method **900** comprises running an electric submersible pump (ESP) assembly into a wellbore, wherein the ESP assembly

comprises an electric motor, a gas separator assembly having a fluid inlet and one or more liquid phase discharge ports (e.g., (A) a single set of one or more liquid phase discharge ports associated with a single cross-over or (B) two sets of one or more liquid phase discharge ports, where each set of liquid phase discharge ports is associated with a different cross-over, as for example in a tandem gas separator configuration), and a centrifugal pump assembly having a fluid inlet fluidically coupled to the liquid discharge port of the gas separator assembly. In an embodiment, the method **900** may be practiced with a tandem gas separator assembly in place of the single gas separator assembly described here with reference to block **902**. A tandem gas separator assembly is illustrated in FIG. **9** and described further below.

(58) At block **904**, the method **900** comprises turning a drive shaft of the gas separator assembly by an electric motor of the ESP assembly. At block **906**, the method **900** comprises drawing reservoir fluid from the wellbore into the gas separator assembly by a first fluid mover of the gas separator assembly that is coupled to the drive shaft. At block **908**, the method **900** comprises moving the reservoir fluid downstream by the first fluid mover (e.g., centrifugal pump stages **405A** and **405B**) within the gas separator assembly.

(59) At block **910**, the method **900** comprises filling an annulus within the gas separator assembly with the reservoir fluid, wherein the annulus is defined between an inside surface of the separator assembly and an outside surface of the drive shaft and wherein the annulus is located downstream of the first fluid mover. In an embodiment, the annulus is provided by the fluid reservoir **170** and may be defined between the drive shaft **172** and an inside surface of the housing **312** or by an inside surface of a sleeve retained by the housing **312**. In an embodiment, a volume of the annulus is at least 50 cubic inches and less than 1000 cubic inches. At block **912**, the method **900** comprises flowing the reservoir fluid from the annulus within the gas separator assembly to a second fluid mover of the gas separator assembly, wherein the second fluid mover is located downstream of the annulus. In an embodiment, the second fluid mover may be the stationary auger **302**. In an embodiment, the second fluid mover may be the paddle wheel **327**. In an embodiment, the second fluid mover may be an impeller without a diffuser.

(60) At block **914**, the method **900** comprises moving the reservoir fluid downstream by the second fluid mover to a gas flow path and liquid flow path separator (e.g., the crossover **350**) of the gas separator assembly. In an embodiment, the processing of block **914** comprises inducing a rotating motion in the reservoir fluid by the second fluid mover and flowing the reservoir fluid into a separation chamber located downstream of the second fluid mover and upstream of the gas flow path and liquid flow path separator. In an embodiment, the processing of block **914** comprises separating gas phase fluid out from liquid phase fluid in the separation chamber by the rotating motion of the reservoir fluid. At block **916**, the method **900** comprises discharging a portion of the reservoir fluid via a gas phase discharge port of the gas flow path and liquid flow path separator to an exterior of the gas separator assembly.

(61) At block **918**, the method **900** comprises discharging a portion of the reservoir fluid via a liquid phase discharge port of the by the gas flow path and liquid flow path separator downstream of the gas separator assembly to the centrifugal pump assembly.

(62) At block **920**, the method **900** comprises pumping the portion of the reservoir fluid discharged via the liquid phase discharge port by the centrifugal pump assembly. At block **922**, the method **900** comprises flowing the portion of the reservoir fluid discharged via the liquid phase discharge port out a discharge of the centrifugal pump assembly via a production tubing to a surface location.

(63) In an embodiment, the method **900** further comprises drawing gas from the wellbore into the gas separator by the first fluid mover; flowing the gas downstream by the first fluid mover within the gas separator assembly; mixing the gas with reservoir fluid retained by the annulus to form a mix of gas and fluid; and flowing the mix of gas and fluid from the annulus within the gas separator assembly to the second fluid mover of the gas separator assembly. In an embodiment, the method **900** further comprises stabilizing the drive shaft by a spider bearing that is concentric with

the drive shaft and that is located inside the annulus within the gas separator assembly, wherein the spider bearing provides flow paths for the reservoir fluid between struts of the spider bearing. In an embodiment, the method **900** comprises stabilizing the drive shaft by a plurality of spider bearings, wherein each spider bearing is concentric with the drive shaft, is located inside the annulus within the gas separator assembly, and provides flow paths for the reservoir fluid between struts of the spider bearing. The plurality of spider bearings may be separated from each other by at least 4 inches and less than 16 inches, at least 6 inches and less than 14 inches, or at least 8 inches and less than 12 inches.

(64) Turning now to FIG. **8A** and FIG. **8B**, a method **950** is described. In an embodiment, the method **950** is a method of assembling an electric submersible pump (ESP) assembly at a wellbore location. At block **952**, the method **950** comprises coupling a downstream end of an electric motor to an upstream end of a seal unit. At block **954**, the method **950** comprises lowering the electric motor, and seal unit partially into the wellbore.

(65) At block **956**, the method **950** comprises coupling a downstream end of the seal unit to an upstream end of a gas separator assembly, wherein the gas separator assembly comprises a drive shaft; a fluid reservoir concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the fluid reservoir and an outside surface of the drive shaft define a first annulus that is fluidically coupled to the fluid outlet of the first fluid mover; a second fluid mover having a fluid inlet and a fluid outlet, wherein the second fluid mover is located downstream of the fluid reservoir, and wherein the fluid inlet of the second fluid mover is fluidically coupled to the first annulus; a separation chamber concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the separation chamber and the outside surface of the drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the second fluid mover; and a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port, wherein the gas flow path and liquid flow path separator has a fluid inlet that is fluidically coupled to the second annulus.

(66) At block **958**, the method **950** comprises lowering the electric motor, seal unit, and gas separator assembly partially into the wellbore. In an embodiment, the gas separator assembly comprises a spider bearing concentric with the drive shaft and located within the first fluid reservoir, wherein the spider bearing comprises struts that provide fluid communication paths between the struts. In an embodiment, the gas separator assembly comprises a plurality of spider bearings concentric with the drive shaft and located within the first fluid reservoir, wherein the spider bearings each comprises struts that provide fluid communication paths between the struts. In an embodiment, the gas separator assembly comprises a plurality of fluid reservoirs. In an embodiment, the gas separator assembly comprises a second fluid reservoir concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the second fluid reservoir and an outside surface of the drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the second fluid mover, wherein the second fluid mover is mechanically coupled to the drive shaft, and comprises a third fluid mover having a fluid inlet and a fluid outlet, wherein the third fluid mover is located downstream of the second fluid reservoir, and wherein the fluid inlet of the third fluid reservoir is fluidically coupled to the second fluid reservoir, wherein the separation chamber and the gas flow path and liquid flow path separator are located downstream of the third fluid mover, wherein the upstream end of the separation chamber is fluidically coupled to the fluid outlet of the third fluid mover, and wherein the fluid inlet of the separation chamber is fluidically coupled to the fluid outlet of the second fluid mover via the third fluid mover and via the second fluid reservoir.

(67) At block **960**, the method **950** comprises coupling a downstream end of the gas separator assembly to an upstream end of centrifugal pump assembly. At block **962**, the method **950** comprises lowering the electric motor, seal unit, gas separator assembly, and centrifugal pump

assembly partially into the wellbore.

(68) Turning now to FIG. 9, a tandem gas separator configuration of the gas separator assembly **126** is described. A tandem gas separator assembly comprises two gas separator assemblies where an upstream gas separator assembly discharges liquid phase fluid out its crossover directly into the inlet of the first fluid mover of the downstream separator assembly, for example directly into an inlet of an impeller of a centrifugal pump stage. The upstream gas separator assembly has its own crossover and the downstream gas separator assembly has its own crossover. In an embodiment, a tandem gas separator assembly may be used to deliver a more liquid rich (e.g., lower gas fluid ratio) to the centrifugal pump assembly **128** by separating gas twice from the reservoir fluid **142**. In an embodiment, since some of the reservoir fluid **142** (e.g., a gas phase rich fraction) is exhausted, the rate of flow of fluid into the downstream gas separator is inherently less than the rate of flow of fluid into the upstream gas separator. In an embodiment, the fluid movers of the upstream gas separator may be designed for a higher rate of flow of fluid, and the downstream gas separator may be designed for a lower rate of flow of fluid.

(69) Much of the tandem gas separator assembly **126** illustrated in FIG. 9 is composed of components described above with reference to FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, and FIG. 3. The tandem gas separator assembly **126** comprises a single base **410** have inlet ports **136**. The upstream gas separator assembly comprises a centrifugal pump **405**, a first fluid reservoir **170A**, a stationary auger **302**, a first separation chamber **303A**, and a crossover **350**. In an embodiment, the centrifugal pump **405** may be replaced in the upstream gas separator by an auger mechanically coupled to the drive shaft **172** or by a paddle wheel **327**. In an embodiment, the stationary auger **302** may be replaced by a paddle wheel **327** mechanically coupled to the drive shaft **172** or an impeller mechanically coupled to the drive shaft **172**. A first gas phase fluid **426A** is discharged by the gas phase discharge **314** of the upstream gas separator into the annulus **210**, and a first liquid phase fluid **428A** is discharged by the liquid phase discharge **316** into the inlet of the centrifugal pump **425** of the downstream gas separator. Note that there is no base having inlet ports between the crossover **350** of the upstream gas separator and the centrifugal pump **425** of the downstream gas separator.

(70) The downstream gas separator of the tandem gas separator assembly **126** comprises a centrifugal pump **425**, a second fluid reservoir **170B**, a paddle wheel **327**, a second separation chamber **303B**, and a crossover **350**. In an embodiment, the centrifugal pump **425** may be replaced in the downstream gas separator by an auger mechanically coupled to the drive shaft **172**. In an embodiment, the paddle wheel **327** may be replaced by a stationary auger **302** or by an impeller mechanically coupled to the drive shaft **172**. A second gas phase fluid **426B** is discharged by the gas phase discharge **314** of the downstream gas separator into the annulus **210**, and a second liquid phase fluid **428B** is discharged by the liquid phase discharge **316** to the centrifugal pump assembly **128**.

ADDITIONAL DISCLOSURE

(71) The following are non-limiting, specific embodiments in accordance with the present disclosure:

(72) A first embodiment, which is a downhole gas separator assembly, comprising a drive shaft, a first fluid mover mechanically coupled to the drive shaft and having a fluid inlet and a fluid outlet, a fluid reservoir concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the fluid reservoir and an outside surface of the drive shaft define a first annulus that is fluidically coupled to the fluid outlet of the first fluid mover, a second fluid mover having a fluid inlet and a fluid outlet, wherein the second fluid mover is located downstream of the fluid reservoir, and wherein the fluid inlet of the second fluid mover is fluidically coupled to the first annulus, a separation chamber concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the separation chamber and the outside surface of the drive shaft define a second annulus that is

fluidically coupled to the fluid outlet of the second fluid mover, and a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port, wherein the gas flow path and liquid flow path separator has a fluid inlet that is fluidically coupled to the second annulus.

(73) A second embodiment, which is the downhole gas separator assembly of the first embodiment, wherein the first annulus has a volume of at least 18 cubic inches and less than 1000 cubic inches.

(74) A third embodiment, which is the downhole gas separator assembly of any of the first and the second embodiments, wherein a distance between the fluid inlet of the first fluid mover and the gas phase discharge port of the gas flow path and liquid flow path separator is at least 4 feet and less than 500 feet.

(75) A fourth embodiment, which is the downhole gas separator assembly of any of the first through the third embodiments, wherein the fluid reservoir is at least 6 inches long and less than 17 inches long.

(76) A fifth embodiment, which is the downhole gas separator assembly of any of the first through the fourth embodiments, further comprising a spider bearing located within the fluid reservoir that has a central through-hole that surrounds the drive shaft.

(77) A sixth embodiment, which is the downhole gas separator assembly of the fifth embodiment, wherein the fluid reservoir is at least 17 inches long and less than 34 inches long.

(78) A seventh embodiment, which is the downhole gas separator assembly of any of the first through the sixth embodiments, further comprising a housing, wherein the inside surface of the fluid reservoir and the inside surface of the separation chamber is provided by an inside surface of the housing, wherein the first fluid mover and the second fluid mover are located within the housing, and wherein the gas flow path and liquid flow path separator is mechanically coupled to the housing.

(79) An eighth embodiment, which is the downhole gas separator assembly of any of the first through the seventh embodiments, further comprising a housing, wherein the inside surface of the separation chamber is provided by an inside surface of the housing, wherein the inside surface of the fluid reservoir is provided by a sleeve that is retained within the housing, wherein the first fluid mover and the second fluid mover are located within the housing, and wherein the gas flow path and liquid flow path separator is mechanically coupled to the housing.

(80) A ninth embodiment, which is the downhole gas separator assembly of any of the first through the eighth embodiments, wherein the second fluid mover is a stationary auger, an auger mechanically coupled to the drive shaft, an impeller mechanically coupled to the drive shaft, a centrifuge rotor mechanically coupled to the drive shaft, or a paddle wheel mechanically coupled to the drive shaft.

(81) A tenth embodiment, which is the downhole gas separator assembly of any of the first through the ninth embodiments, wherein the first fluid mover is a centrifugal pump having at least one centrifugal pump stage wherein each centrifugal pump stage comprises an impeller mechanically coupled to the drive shaft and a diffuser.

(82) An eleventh embodiment, which is the downhole gas separator assembly of any of the first through the tenth embodiments, wherein the second fluid mover is mechanically coupled to the drive shaft and further comprising a second fluid reservoir concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the second fluid reservoir and an outside surface of the drive shaft define a third annulus that is fluidically coupled to the fluid outlet of the second fluid mover, and a third fluid mover having a fluid inlet and a fluid outlet, wherein the third fluid mover is located downstream of the second fluid reservoir and is located upstream of the separation chamber, wherein the fluid inlet of the third fluid mover is fluidically coupled to the third annulus, and wherein the fluid outlet of the third fluid mover is fluidically coupled to the second annulus.

(83) A twelfth embodiment, which is the downhole gas separator assembly of any of the first

through the eleventh embodiments, further comprising a base having an inlet, a fourth fluid mover mechanically coupled to the drive shaft, located upstream of the base, having a fluid outlet, and having a fluid inlet fluidically coupled to the inlet of the base, a third fluid reservoir concentrically disposed around the drive shaft and located downstream of the fourth fluid mover, wherein an inside surface of the third fluid reservoir and the outside surface of the drive shaft define a fourth annulus that is fluidically coupled to the fluid outlet of the fourth fluid mover, a fifth fluid mover having a fluid inlet and a fluid outlet, wherein the fifth fluid mover is located downstream of the third fluid reservoir, and wherein the fluid inlet of the fifth fluid mover is fluidically coupled to the fourth annulus, a second separation chamber concentrically disposed around the drive shaft and located downstream of the fifth fluid mover, wherein an inside surface of the second separation chamber and the outside surface of the drive shaft define a fifth annulus that is fluidically coupled to the fluid outlet of the fifth fluid mover, and a second gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port fluidically coupled to the fluid inlet of the first fluid mover, and wherein the gas flow path and liquid flow path separator has a fluid inlet fluidically coupled to the fifth annulus.

(84) A thirteenth embodiment, which is a method of lifting liquid in a wellbore, comprising running an electric submersible pump (ESP) assembly into a wellbore, wherein the ESP assembly comprises an electric motor, a gas separator assembly having a fluid inlet and a liquid phase discharge port, and a centrifugal pump assembly having a fluid inlet fluidically coupled to the liquid discharge port of the gas separator assembly, turning a drive shaft of the gas separator assembly by an electric motor of the ESP assembly, drawing reservoir fluid from the wellbore into the gas separator assembly by a first fluid mover of the gas separator assembly that is coupled to the drive shaft, moving the reservoir fluid downstream by the first fluid mover within the gas separator assembly, filling an annulus within the gas separator assembly with the reservoir fluid, wherein the annulus is defined between an inside surface of the gas separator assembly and an outside surface of the drive shaft and wherein the annulus is located downstream of the first fluid mover, flowing the reservoir fluid from the annulus within the gas separator assembly to a second fluid mover of the gas separator, wherein the second fluid mover is located downstream of the annulus, moving the reservoir fluid downstream by the second fluid mover to a gas flow path and liquid flow path separator of the gas separator assembly, discharging a portion of the reservoir fluid via a gas phase discharge port of the gas flow path and liquid flow path separator to an exterior of the gas separator assembly, discharging a portion of the reservoir fluid via a liquid phase discharge port of the gas flow path and liquid flow path separator downstream of the gas separator assembly to the centrifugal pump assembly, pumping the portion of the reservoir fluid discharged via the liquid phase discharge port by the centrifugal pump assembly, and flowing the portion of the reservoir fluid discharged via the liquid phase discharge port out a discharge of the centrifugal pump assembly via a production tubing to a surface location.

(85) A fourteenth embodiment, which is the method of the thirteenth embodiment further comprising drawing gas from the wellbore into the gas separator by the first fluid mover, flowing the gas downstream by the first fluid mover within the gas separator assembly, mixing the gas with reservoir fluid retained by the annulus to form a mix of gas and fluid, and flowing the mix of gas and fluid from the annulus within the gas separator assembly to the second fluid mover of the gas separator assembly.

(86) A fifteenth embodiment, which is the method of the fourteenth embodiment, wherein a volume of the annulus is at least 50 cubic inches and less than 1000 cubic inches.

(87) A sixteenth embodiment, which is the method of the twelfth embodiment, further comprising stabilizing the drive shaft by a spider bearing that is concentric with the drive shaft and that is located inside the annulus within the gas separator assembly, wherein the spider bearing provides flow paths for the reservoir fluid between struts of the spider bearing.

(88) A seventeenth embodiment, which is the method of the twelfth embodiment, further

comprising stabilizing the drive shaft by a plurality of spider bearings, wherein each spider bearing is concentric with the drive shaft, is located inside the annulus within the gas separator assembly, and provides flow paths for the reservoir fluid between struts of the spider bearing.

(89) An eighteenth embodiment, which is the method of the seventeenth embodiment, wherein each spider bearing is separated from the other spider bearing by at least 4 inches and less than 16 inches.

(90) A nineteenth embodiment, which is a method of assembling an electric submersible pump (ESP) assembly at a wellbore location, comprising coupling a downstream end of an electric motor to an upstream end of a seal unit, lowering the electric motor, and seal unit partially into the wellbore, coupling a downstream end of the seal unit to an upstream end of a gas separator assembly, wherein the gas separator assembly comprises a drive shaft, a first fluid mover mechanically coupled to the drive shaft and having a fluid inlet and a fluid outlet, a fluid reservoir concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the fluid reservoir and an outside surface of the drive shaft define a first annulus that is fluidically coupled to the fluid outlet of the first fluid mover, a second fluid mover having a fluid inlet and a fluid outlet, wherein the second fluid mover is located downstream of the fluid reservoir, and wherein the fluid inlet of the second fluid mover is fluidically coupled to the first annulus, a separation chamber concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the separation chamber and the outside surface of the drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the second fluid mover, and a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port, wherein the gas flow path and liquid flow path separator has a fluid inlet that is fluidically coupled to the second annulus, lowering the electric motor, seal unit, and gas separator assembly partially into the wellbore, coupling a downstream end of the gas separator assembly to an upstream end of centrifugal pump assembly, and lowering the electric motor, seal unit, gas separator assembly, and centrifugal pump assembly partially into the wellbore.

(91) A twentieth embodiment, which is the method of the nineteenth embodiment, wherein the gas separator assembly comprises a plurality of fluid reservoirs.

(92) A twenty-first embodiment, which is the method of any of the nineteenth and the twentieth embodiments, wherein the second fluid mover is mechanically coupled to the drive shaft and gas separator assembly comprises a second fluid reservoir concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the second fluid reservoir and an outside surface of the drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the second fluid mover, and a third fluid mover having a fluid inlet and a fluid outlet, wherein the third fluid mover is located downstream of the second fluid reservoir, and wherein the fluid inlet of the third fluid reservoir is fluidically coupled to the second fluid reservoir, wherein the gas flow path and liquid flow path separator is located downstream of the third fluid mover, wherein the fluid inlet of the gas flow path and liquid flow path separator is fluidically coupled to the fluid outlet of the third fluid mover, and wherein the fluid inlet of the gas flow path and liquid flow path separator is fluidically coupled to the fluid outlet of the second fluid mover via the third fluid mover and via the second fluid reservoir.

(93) A twenty-second embodiment, which is the method of any of the nineteen through the twenty-first embodiments, wherein the gas separator assembly further comprises a spider bearing concentric with the drive shaft and located within the first fluid reservoir, wherein the spider bearing comprises struts that provide fluid communication paths between the struts.

(94) While embodiments have been shown and described, modifications thereof can be made by one skilled in the art without departing from the spirit and teachings of this disclosure. The embodiments described herein are exemplary only, and are not intended to be limiting. Many variations and modifications of the embodiments disclosed herein are possible and are within the

scope of this disclosure. Where numerical ranges or limitations are expressly stated, such express ranges or limitations should be understood to include iterative ranges or limitations of like magnitude falling within the expressly stated ranges or limitations (e.g., from about 1 to about 10 includes, 2, 3, 4, etc.; greater than 0.10 includes 0.11, 0.12, 0.13, etc.). For example, whenever a numerical range with a lower limit, R_l , and an upper limit, R_u , is disclosed, any number falling within the range is specifically disclosed. In particular, the following numbers within the range are specifically disclosed: $R = R_l + k \cdot (R_u - R_l)$, wherein k is a variable ranging from 1 percent to 100 percent with a 1 percent increment, i.e., k is 1 percent, 2 percent, 3 percent, 4 percent, 5 percent, . . . 50 percent, 51 percent, 52 percent, . . . , 95 percent, 96 percent, 97 percent, 98 percent, 99 percent, or 100 percent. Moreover, any numerical range defined by two R numbers as defined in the above is also specifically disclosed. Use of the term “optionally” with respect to any element of a claim is intended to mean that the subject element is required, or alternatively, is not required. Both alternatives are intended to be within the scope of the claim. Use of broader terms such as comprises, includes, having, etc. should be understood to provide support for narrower terms such as consisting of, consisting essentially of, comprised substantially of, etc.

(95) Accordingly, the scope of protection is not limited by the description set out above but is only limited by the claims which follow, that scope including all equivalents of the subject matter of the claims. Each and every claim is incorporated into the specification as an embodiment of the present disclosure. Thus, the claims are a further description and are an addition to the embodiments of the present disclosure. The discussion of a reference herein is not an admission that it is prior art, especially any reference that may have a publication date after the priority date of this application. The disclosures of all patents, patent applications, and publications cited herein are hereby incorporated by reference, to the extent that they provide exemplary, procedural, or other details supplementary to those set forth herein.

Claims

1. A downhole gas separator assembly, comprising: a drive shaft; a first fluid reservoir concentrically disposed around the drive shaft wherein an inside surface of the first fluid reservoir and an outside surface of the drive shaft define a first annulus within a portion of the first fluid reservoir that does not enclose a radial support of the drive shaft, wherein when the separator drive shaft is about 0.6875 inches in diameter, the first annulus has a volume of at least 70 cubic inches and less than 100 cubic inches; wherein when the separator drive shaft is about 0.875 inches in diameter, the first annulus has a volume of at least 85 cubic inches and less than 120 cubic inches; wherein when the separator drive shaft is about 1.0 inches in diameter, the first annulus has a volume of at least 180 cubic inches and less than 250 cubic inches; and wherein when the separator drive shaft is about 1.1875 inches in diameter, the first annulus has a volume of at least 220 cubic inches and less than 300 cubic inches; a first fluid mover having a fluid inlet and a fluid outlet, wherein the first fluid mover is located downstream of the first fluid reservoir, and wherein the fluid inlet of the first fluid mover is fluidically coupled to the first annulus; a separation chamber concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the separation chamber and the outside surface of the drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the first fluid mover; and a crossover defining a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port, wherein the gas flow path and liquid flow path separator has a fluid inlet that is fluidically coupled to the second annulus.
2. The downhole gas separator assembly of claim 1, wherein a distance between a downhole end of the downhole gas separator assembly and the gas phase discharge port of the crossover is at least 8 feet and less than 500 feet.
3. The downhole gas separator assembly of claim 1, wherein a distance between a downhole end of

the downhole gas separator assembly and the gas phase discharge port of the crossover is at least 14 feet and less than 500 feet.

4. The downhole gas separator assembly of claim 1, wherein a distance between a downhole end of the downhole gas separator assembly and the gas phase discharge port of the crossover is at least 22 feet and less than 500 feet.

5. The downhole gas separator assembly of claim 1, further comprising a spider bearing comprising a plurality of vanes, wherein the spider bearing is concentric with the drive shaft, the spider bearing is located within the first fluid reservoir, and the spider bearing provides flow paths between the vanes.

6. The downhole gas separator assembly of claim 5, wherein the first fluid reservoir is at least 17 inches long and less than 34 inches long.

7. The downhole gas separator assembly of claim 5, wherein the spider bearing comprises four vanes.

8. The downhole gas separator assembly of claim 5, wherein the spider bearing comprises five vanes.

9. The downhole gas separator assembly of claim 1, wherein the first fluid mover is a stationary auger, an auger mechanically coupled to the drive shaft, an impeller mechanically coupled to the drive shaft, or a paddle wheel mechanically coupled to the drive shaft.

10. The downhole gas separator assembly of claim 1, further comprising a second fluid mover disposed around the drive shaft and located downstream of the first fluid reservoir.

11. The downhole gas separator assembly of claim 10, further comprising: a second fluid reservoir concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the second fluid reservoir and an outside surface of the drive shaft define a third annulus that is fluidically coupled to the fluid outlet of the second fluid mover; and a third fluid mover having a fluid inlet and a fluid outlet, wherein the third fluid mover is located downstream of the second fluid reservoir and is located upstream of the separation chamber, wherein the fluid inlet of the third fluid mover is fluidically coupled to the third annulus, and wherein the fluid outlet of the third fluid mover is fluidically coupled to the second annulus.

12. The downhole gas separator assembly of claim 1, further comprising: a base having an inlet; a second fluid mover mechanically coupled to the drive shaft, located downstream of the base, having a fluid outlet, and having a fluid inlet fluidically coupled to the inlet of the base; a second fluid reservoir concentrically disposed around the drive shaft and located downstream of the second fluid mover, wherein an inside surface of the second fluid reservoir and the outside surface of the drive shaft define a third annulus that is fluidically coupled to the fluid outlet of the second fluid mover; a third fluid mover having a fluid inlet and a fluid outlet, wherein the third fluid mover is located downstream of the second fluid reservoir, and wherein the fluid inlet of the third fluid mover is fluidically coupled to the third annulus; a second separation chamber concentrically disposed around the drive shaft and located downstream of the third fluid mover, wherein an inside surface of the second separation chamber and the outside surface of the drive shaft define a fourth annulus that is fluidically coupled to the fluid outlet of the third fluid mover; and a second crossover defining a gas flow path and liquid flow path separator having a second gas phase discharge port open to an exterior of the assembly and a second liquid phase discharge port fluidically coupled to the fluid inlet of the first fluid reservoir, and wherein the second crossover has a fluid inlet fluidically coupled to the fourth annulus.

13. A method of lifting liquid in a wellbore, comprising: running an electric submersible pump (ESP) assembly into a wellbore, wherein the ESP assembly comprises an electric motor having a motor drive shaft, a seal section having a seal drive shaft coupled to the motor drive shaft, a gas separator assembly having a separator drive shaft coupled to the seal drive shaft, a fluid reservoir concentrically disposed around the separator drive shaft, wherein an inside surface of the fluid reservoir and an outside surface of the separator drive shaft define a first annulus within a portion

of the fluid reservoir that does not enclose a radial support of the drive shaft, wherein when the separator drive shaft is about 0.6875 inches in diameter, the first annulus has a volume of at least 70 cubic inches and less than 100 cubic inches; wherein when the separator drive shaft is about 0.875 inches in diameter, the first annulus has a volume of at least 85 cubic inches and less than 120 cubic inches; wherein when the separator drive shaft is about 1.0 inches in diameter, the first annulus has a volume of at least 180 cubic inches and less than 250 cubic inches; and wherein when the separator drive shaft is about 1.1875 inches in diameter, the first annulus has a volume of at least 220 cubic inches and less than 300 cubic inches, a first fluid mover having a fluid inlet and a fluid outlet, wherein the first fluid mover is located downstream of the fluid reservoir, and wherein the fluid inlet of the first fluid mover is fluidically coupled to the first annulus, a separation chamber concentrically disposed around the separator drive shaft and located downstream of the first fluid mover, wherein an inside surface of the separation chamber and the outside surface of the separator drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the first fluid mover, and a crossover defining a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port, wherein the crossover has a third fluid inlet that is fluidically coupled to the second annulus, and a centrifugal pump assembly having a pump drive shaft coupled to the separator drive shaft and a fluid inlet fluidically coupled to the liquid discharge port of the crossover of the gas separator assembly; turning the separator drive shaft by the electric motor of the ESP assembly; drawing reservoir fluid from the wellbore into the gas separator assembly; filling the first annulus with the reservoir fluid; flowing the reservoir fluid from the first annulus to the first fluid mover moving the reservoir fluid downstream by the first fluid mover to the crossover; discharging at least some of the gas phase fluid portion of the reservoir fluid via the gas phase discharge port of the crossover to an exterior of the gas separator assembly; discharging at least some of the liquid phase portion of the reservoir fluid via the liquid phase discharge port of the crossover to the centrifugal pump assembly; pumping the portion of the reservoir fluid discharged via the liquid phase discharge port by the centrifugal pump assembly; and flowing the portion of the reservoir fluid discharged via the liquid phase discharge port out a discharge of the centrifugal pump assembly via a production tubing to a surface location.

14. The method of claim 13, further comprising: drawing reservoir fluid from the wellbore into the gas separator assembly at a second time, wherein the reservoir fluid at the second time contains only gas phase fluid; mixing the reservoir fluid containing only gas phase fluid with reservoir fluid retained by the first annulus within the gas separator assembly to form a mix of gas phase fluid and liquid phase fluid; and flowing the mix of gas phase fluid and liquid phase fluid from the first annulus within the gas separator assembly to the first fluid mover of the gas separator assembly.

15. The method of claim 13, further comprising stabilizing the drive shaft by a spider bearing comprising a plurality of vanes, wherein the spider bearing is concentric with the drive shaft, the spider bearing is located within the fluid reservoir, and the spider bearing provides flow paths between the vanes.

16. The method of claim 13, further comprising stabilizing the drive shaft by a spider bearing comprising four vanes, wherein the spider bearing is concentric with the drive shaft, the spider bearing is located within the fluid reservoir, and the spider bearing provides flow paths between the vanes.

17. The method of claim 13, further comprising stabilizing the drive shaft by a spider bearing comprising five vanes, wherein the spider bearing is concentric with the drive shaft, the spider bearing is located within the fluid reservoir, and the spider bearing provides flow paths between the vanes.

18. A method of assembling an electric submersible pump (ESP) assembly at a wellbore location, comprising: coupling a downstream end of an electric motor to an upstream end of a seal unit; lowering the electric motor, and seal unit partially into the wellbore; coupling a downstream end of

the seal unit to an upstream end of a gas separator assembly, wherein the gas separator assembly comprises a drive shaft, a fluid reservoir concentrically disposed around the drive shaft, wherein an inside surface of the fluid reservoir and an outside surface of the drive shaft define a first annulus within a portion of the fluid reservoir that does not enclose a radial support of the drive shaft, wherein when the separator drive shaft is about 0.6875 inches in diameter, the first annulus has a volume of at least 70 cubic inches and less than 100 cubic inches; wherein when the separator drive shaft is about 0.875 inches in diameter, the first annulus has a volume of at least 85 cubic inches and less than 120 cubic inches; wherein when the separator drive shaft is about 1.0 inches in diameter, the first annulus has a volume of at least 180 cubic inches and less than 250 cubic inches; and wherein when the separator drive shaft is about 1.1875 inches in diameter, the first annulus has a volume of at least 220 cubic inches and less than 300 cubic inches; a first fluid mover having a fluid inlet and a fluid outlet, wherein the first fluid mover is located downstream of the fluid reservoir, and wherein the fluid inlet of the first fluid mover is fluidically coupled to the first annulus; a separation chamber concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the separation chamber and the outside surface of the drive shaft define a second annulus that is fluidically coupled to the fluid outlet of the first fluid mover; and a crossover defining a gas flow path and liquid flow path separator having a gas phase discharge port open to an exterior of the assembly and a liquid phase discharge port, wherein the gas flow path and liquid flow path separator has a fluid inlet that is fluidically coupled to the second annulus; lowering the electric motor, seal unit, and gas separator assembly partially into the wellbore; coupling a downstream end of the gas separator assembly to an upstream end of a centrifugal pump assembly; and lowering the electric motor, seal unit, gas separator assembly, and centrifugal pump assembly partially into the wellbore.

19. The method of claim 18, wherein the gas separator assembly comprises: a second fluid reservoir concentrically disposed around the drive shaft and located downstream of the first fluid mover, wherein an inside surface of the second fluid reservoir and an outside surface of the drive shaft define a third annulus that is fluidically coupled to the fluid outlet of the first fluid mover; and a second fluid mover having a fluid inlet and a fluid outlet, wherein the second fluid mover is located downstream of the second fluid reservoir, and wherein the fluid inlet of the second fluid mover is fluidically coupled to the second fluid reservoir, wherein the crossover is located downstream of the second fluid mover, wherein the fluid inlet of the crossover is fluidically coupled to the fluid outlet of the second fluid mover, and wherein the fluid inlet of the crossover is fluidically coupled to the fluid outlet of the first fluid mover via the second fluid mover and via the second fluid reservoir.

20. The method of claim 19, wherein the gas separator assembly further comprises a spider bearing concentric with the drive shaft and located within the first fluid reservoir, wherein the spider bearing comprises four vanes or five vanes, and further comprising stabilizing the drive shaft by the spider bearing.
